More often we write like this:



$4^2 = 16$

This says "4 squared is 16"

The 2 at the top stands for squared and it indicates the number of times the number 4 appears in the product ($4^2 = 4 \times 4 = 16$).

The numbers 1, 4, 9, 16, ... are all square numbers (also called perfect square numbers). Each of them is made up of the product of same two factors.

A natural number n is called a square number, if we can find another natural number m such that $n = m^2$.

Is 49 a square number? Yes, because it can be written as 7^2. Is 50 a square number?

The following table gives the squares of numbers up to 20.

Number	Its square	Number	Its square	Number	Its square	Number	Its square
1	1	6	36	11	121	16	256
2	4	7	49	12	144	17	289
3	9	8	64	13	169	18	324
4	16	9	81	14	196	19	361
5	25	10	100	15	225	20	400

Try to extend the table up to 50 numbers!

We can now easily verify the following properties of square numbers by referring the table given above:

- The square numbers end with 0, 1, 4, 5, 6 and 9 only.
- If a number ends with 1 or 9, its square ends with 1.
- If a number ends with 2 or 8, its square ends with 4.
- If a number ends with 3 or 7, its square ends with 9.
- If a number ends with 4 or 6, its square ends with 6.
- If a number ends with 5 or 0, its square also ends with 5 or 0 respectively.
- Square of an odd number is always odd and the square of an even number is always even.
- Numbers that end with 2,3,7 and 8 are not perfect squares.



Think

1. Is the square of a prime number, prime?
2. Will the sum of two perfect squares always be a perfect square? What about their difference and their product?



Try these

1. Which among 256, 576, 960, 1025, 4096 are perfect square numbers? (Hint: Try to extend the table of squares already seen).
2. One can judge just by look that each of the following numbers 82, 113, 1972, 2057, 8888, 24353 is not a perfect square. Explain why?

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